

PF LAB#2 TASKS

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SECTION: BSCS-1K

ROLL NO. : 25K-0516

TASK 1

Scratch - Imagine, Program, Share X PF Lab BCS-1K Section 1K X Scratch - Imagine, Program, Share X Scratch - Imagine, Program, Share X

scratch.mit.edu/projects/editor/?tutorial=getStarted

Scratch Settings File Edit Tutorials Debug Join Scratch Sign in

Code Costumes Sounds

Motion

- move 10 steps
- turn 15 degrees
- turn 15 degrees
- go to random position
- go to x: 0 y: 0
- glide 1 secs to random position
- glide 1 secs to x: 0 y: 0
- point in direction 90
- point towards mouse-pointer
- change x by 10

Events

- when clicked

Control

- ask What's your name? and wait
- say join Hello answer for 1 seconds
- say Welcome to PF class! for 1 seconds

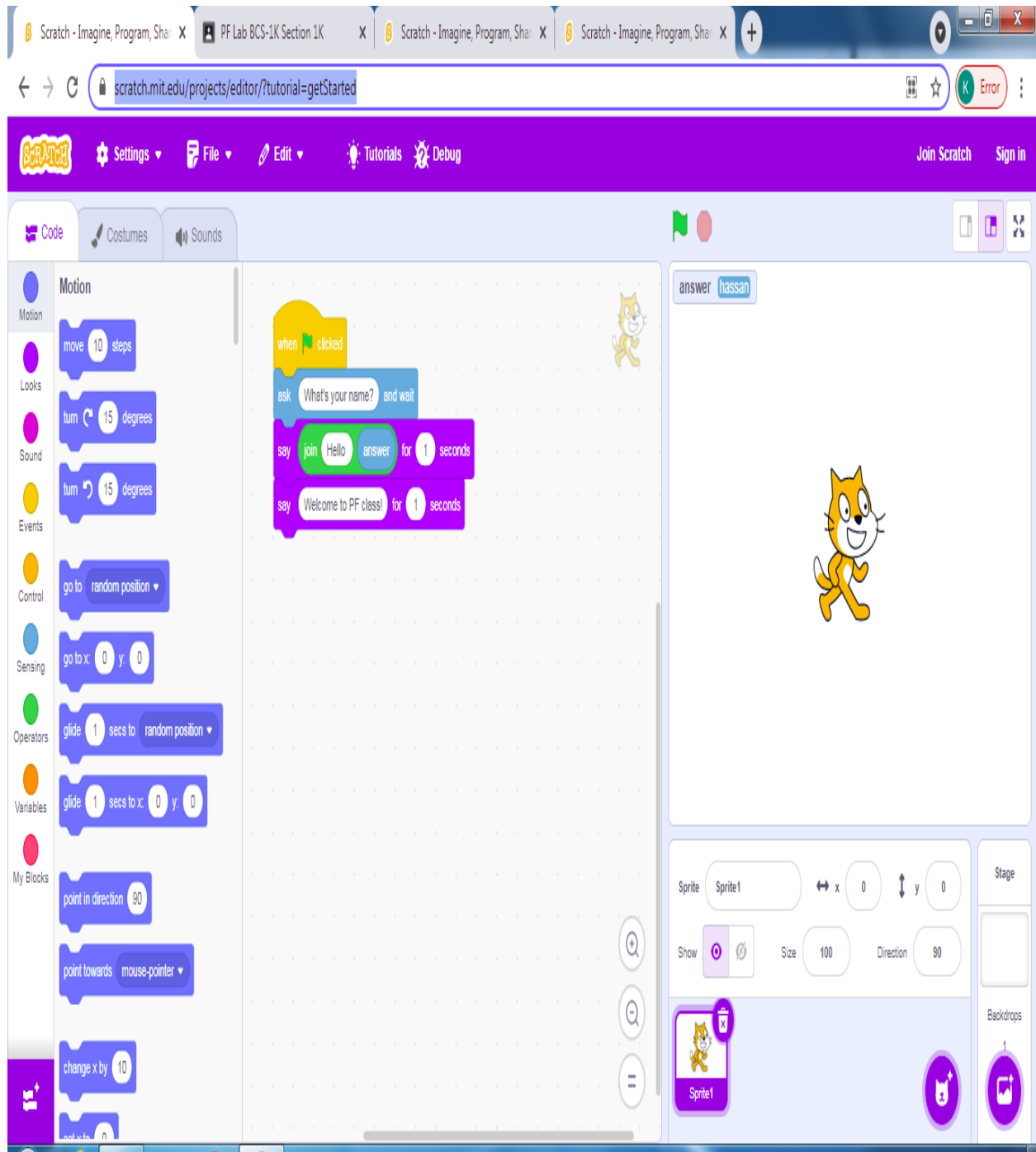
Variables

- answer hassan

Sprite Sprite1 x: 0 y: 0

Stage

Backdrops 1



TASK 2

The image displays two screenshots of the Scratch IDE, showing a project titled "TASK 2". The project is a math application that calculates the sum, difference, product, and quotient of three numbers entered by the user.

Top Screenshot:

- Script Area:** The script starts with a "when clicked" event block, followed by three "ask" blocks to prompt the user for "first number", "second number", and "third number". Each "ask" block is followed by a "wait" block.
- Variables Area:** The "Variables" panel shows the following variables: "difference" (set to 0), "num1" (set to 2), "num2" (set to 4), "num3" (set to 6), "sum" (set to 12), "product" (set to 48), and "quotient" (set to 3).
- Stage Area:** The stage shows the Scratch cat sprite and the current values of the variables: num1: 2, answer: 6, num2: 4, num3: 6, sum: 12, difference: 4, product: 48, and quotient: 3.

Bottom Screenshot:

- Script Area:** The script is updated to include "set" blocks for "num2" and "num3" after the "ask" blocks. It also includes "say" blocks to display the results: "the sum is", "the difference is", "the product is", and "the quotient is".
- Variables Area:** The "Variables" panel shows the same variables as the top screenshot, but the "sum" variable is now set to 12, and the "product" and "quotient" variables are set to 48 and 3, respectively.
- Stage Area:** The stage shows the Scratch cat sprite and the current values of the variables: num1: 2, answer: 6, num2: 4, num3: 6, sum: 12, difference: 4, product: 48, and quotient: 3.

TASK 3

The screenshot shows the Scratch IDE interface. The top bar includes the Scratch logo, settings, file, edit, tutorials, and debug menus. The main workspace is divided into three panels: Variables, Code, and Sprites. The Variables panel on the left shows a list of variables: area, length, width, and my variable. The Code panel in the center contains a script starting with a 'when clicked' event, followed by two 'ask' blocks to prompt the user for length and width, then 'set' blocks to store the input, a calculation block for area, and two 'say' blocks to display the result. The Sprites panel on the right shows the Scratch cat sprite on the stage with a speech bubble saying '15'. The stage also displays the variables 'length', 'answer', 'width', and 'area' with their current values.

Scratch - Imagine, Program, Share X PF Lab BCS-1K Section 1K X Scratch - Imagine, Program, Share X Scratch - Imagine, Program, Share X

scratch.mit.edu/projects/editor/?tutorial=getStarted

Scratch Settings File Edit Tutorials Debug Join Scratch Sign in

Code Costumes Sounds

Variables

Make a Variable

area

length

my variable

width

set area to 0

change area by 1

show variable area

hide variable area

Make a List

My Blocks

Make a Block

when clicked

ask Enter the length and wait

set length to answer

ask Enter the width and wait

set width to answer

set area to length * width

say Area of rectangle is for 2 seconds

say area

length 3

answer 5

width 5

area 15

15

Sprite Sprite1 x 0 y 0

Show Size 100 Direction 90

Stage

Backdrops

TASK 4

The screenshot shows the Scratch IDE interface. The browser address bar displays `scratch.mit.edu/projects/editor/?tutorial=getStarted`. The Scratch logo and navigation buttons (Settings, File, Edit, Tutorials, Debug) are visible in the top purple bar. The left sidebar shows the 'Looks' category selected, with various blocks like 'say', 'think', 'switch costume', 'next costume', 'switch backdrop', and 'next backdrop' visible. The main workspace contains a script for a cat sprite:

```
when clicked
ask Enter your age in years and wait
set age in years to answer
set age in months to 12 * age in years
say Your age in months is for 2 seconds
say age in months for 2 seconds
```

The stage area shows a cat sprite with the following input fields:

- age in years: 4
- answer: 4
- age in months: 48

The bottom right panel shows the 'Sprite' section with 'Sprite1' selected, and the 'Stage' section with 'Backdrops' visible.

TASK 5

PF Lab BCS-1K Section 1K X Scratch - Imagine, Program, Share X

scratch.mit.edu/projects/editor/?tutorial=getStarted

Settings File Edit Tutorials Debug Join Scratch Sign in

Code Costumes Sounds

Looks

say Hello! for 2 seconds

say Hello!

think Hmm... for 2 seconds

think Hmm...

switch costume to costume2

next costume

switch backdrop to backdrop1

next backdrop

change size by 10

set size to 100 %

when clicked

ask Enter your weight and wait

set weight to answer

ask Enter your height and wait

set height to answer

set HEIGHT to height * height

set BMI to weight / HEIGHT

if BMI > 25 then

say BMI is greater than 25 for 1 seconds

else

think BMI is less than 25 for 1 seconds

weight 70

answer 172

height 172

HEIGHT 29584

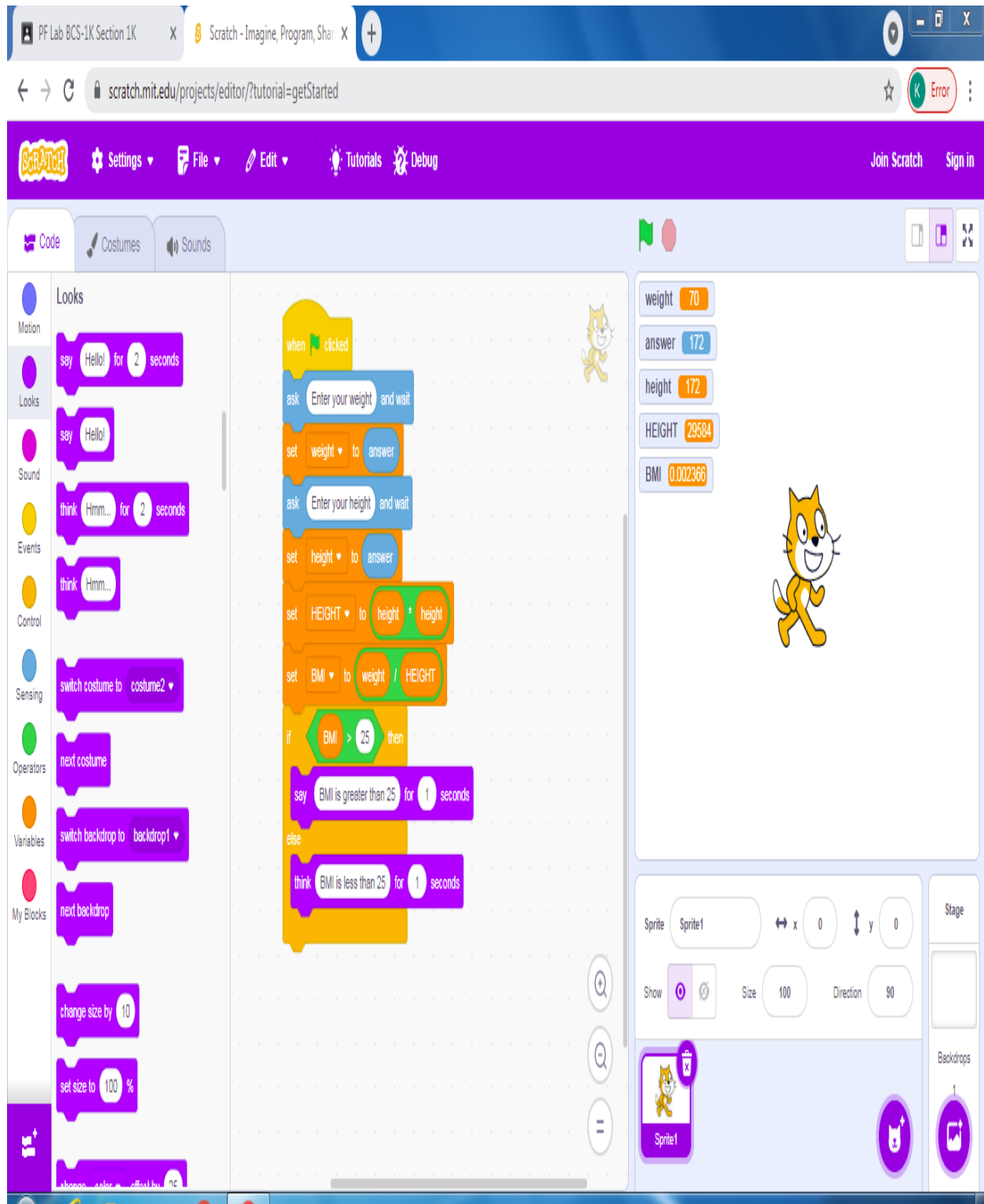
BMI 0.002366

Sprite1

Size 100

Direction 90

Backdrops



TASK 6

PF Lab BCS-1K Section 1K X Scratch - Imagine, Program, Share X Scratch - Imagine, Program, Share X +

scratch.mit.edu/projects/editor/?tutorial=getStarted

Scratch Settings File Edit Tutorials Debug Join Scratch Sign in

Code Costumes Sounds

Control

wait 1 seconds

repeat 10

forever

Sensing

Operators

if then

Variables

if then else

wait until

repeat until

when clicked

ask Enter a number and wait

set number to answer

set counter to 1

repeat until counter = 10

say number + counter for 1 seconds

change counter by 1

number 5

answer 5

counter 10

Sprite Sprite1 x 0 y 0

Show Size 100 Direction 90

Stage

Backdrops

TASK 7

PF Lab BCS-1K Section 1K X Scratch - Imagine, Program, Share X

scratch.mit.edu/projects/editor/?tutorial=getStarted

Settings File Edit Tutorials Debug Join Scratch Sign in

Code Costumes Sounds

Looks

say Hello! for 2 seconds

say Hello!

think Hmm... for 2 seconds

think Hmm...

switch costume to costume2

next costume

switch backdrop to backdrop1

next backdrop

change size by 10

set size to 100 %

when clicked

ask enter time and wait

set time to answer

repeat time

wait 1 seconds

change time by -1

say Time's up! for 2 seconds

time 0

answer 6

Sprite1

Size 100 Direction 90

Stage

Backdrops

TASK 8

PF Lab BCS-1K Section 1K X Scratch - Imagine, Program, Share X Scratch - Imagine, Program, Share X Scratch - Imagine, Program, Share X +

← → ↻ scratch.mit.edu/projects/editor/?tutorial=getStarted ☆ Error

Scratch Settings File Edit Tutorials Debug Join Scratch Sign in

Code Costumes Sounds

Motion: pick random 1 to 10

Looks: > 50 < 50 = 50

Sound: > 50 < 50 = 50

Events: when clicked

Control: ask Enter a year and wait set year to answer

Sensing: if year mod 4 = 0 or year mod 400 = 0 then

Operators: and or not

Variables: join apple banana letter 1 of apple length of apple apple contains a ?

My Blocks: mod round

year 2008
answer 2008

when clicked
ask Enter a year and wait
set year to answer
if year mod 4 = 0 or year mod 400 = 0 then
say It is a leap year for 2 seconds
else
say It is not a leap year for 2 seconds

Sprite: Sprite1 x 0 y 0
Show Size 100 Direction 90
Stage
Backdrops 1

TASK 9

PF Lab BCS-1K Section 1K X Scratch - Imagine, Program, Share X Scratch - Imagine, Program, Share X +

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Scratch Settings File Edit Tutorials Debug Join Scratch Sign in

Code Costumes Sounds

Variables

- Motion
- Looks
- Sound
- Events
- Control
- Sensing
- Operators
- Variables
 - Make a Variable
 - ☒ counter
 - ☐ my variable
 - ☒ number
- My Blocks
 - Make a List
 - Make a Block

My Blocks

when clicked

ask Enter a number and wait

set number to answer

set counter to 1

repeat until counter = number

wait 1 seconds

change counter by 1

say Countdown completed! for 2 seconds

number 10

answer 10

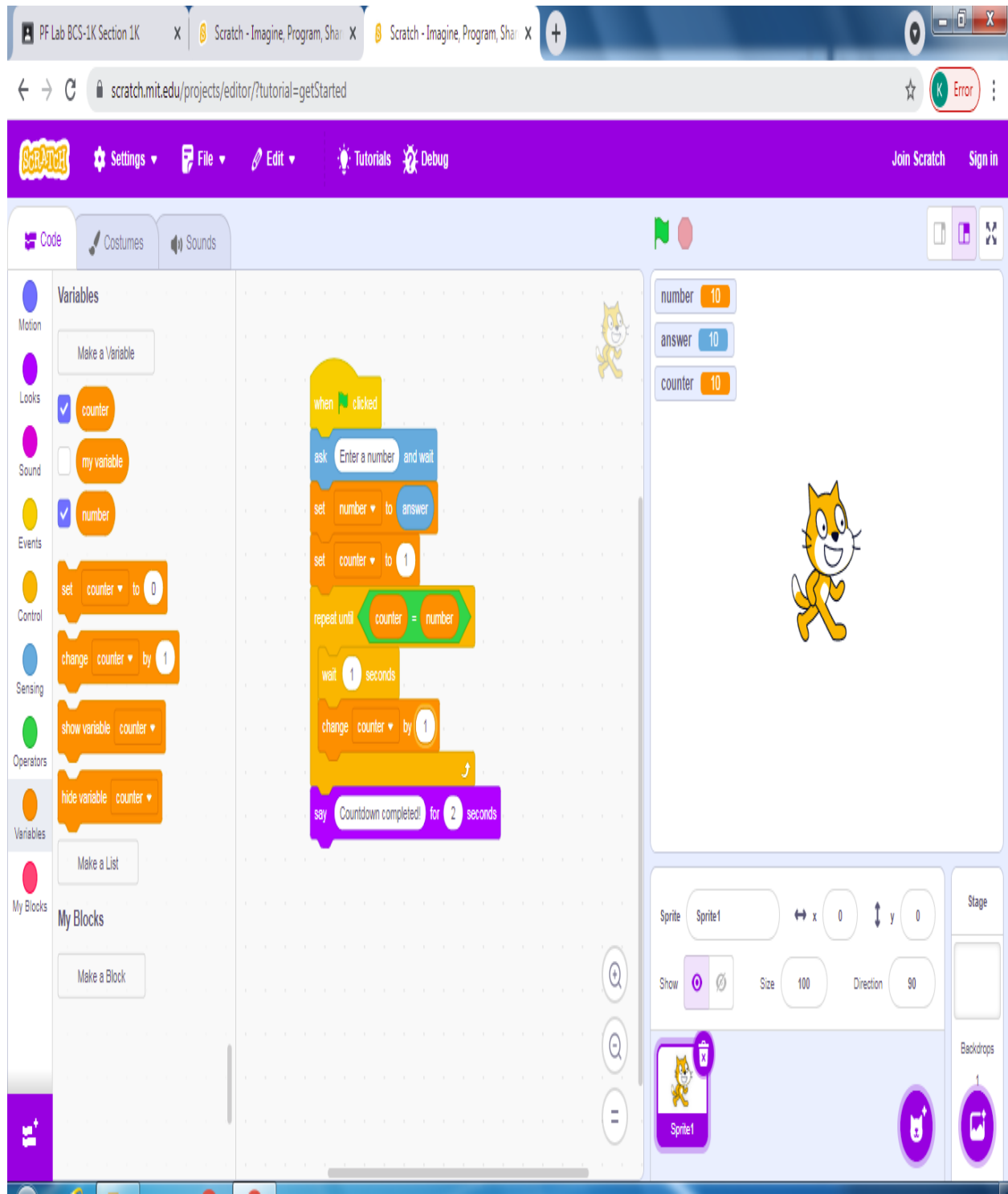
counter 10

Sprite1

Sprite1

Stage

Backdrops



TASK 10

PF Lab BCS-1K Section 1K X Scratch - Imagine, Program, Share X Scratch - Imagine, Program, Share X Scratch - Imagine, Program, Share X

scratch.mit.edu/projects/editor/?tutorial=getStarted

Settings File Edit Tutorials Debug Join Scratch Sign in

Code Costumes Sounds

Variables

Motion

Looks

Sound

Events

Control

Sensing

Operators

Variables

My Blocks

Make a Variable

my variable

p

PRT

r

SI

t

set my variable to 0

change my variable by 1

show variable my variable

hide variable my variable

Make a List

My Blocks

Make a Block

when green flag clicked

ask Enter the principle amount and wait

set p to answer

ask Enter the rate and wait

set r to answer

ask Enter the time and wait

set t to answer

set PRT to $p * r * t$

set SI to $PRT / 100$

if $t < 5$ then

set r to 2.5

else

set r to 10

say The simple interest is for 1 seconds

say SI for 1 seconds

p 30

answer 8

r 10

t 8

PRT 3120

SI 31.2

Sprite1

Stage

Backdrops